RTFF: Random-to-Target Fabric Flattening Policy using Dual-Arm Manipulator

Kai Tang*, Dipankar Bhattacharya*, Hang Xu, Fuyuki Tokuda, Norman C. Tien, and Kazuhiro Kosuge

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Scan the QR to access the Homepage, arXiv and code:

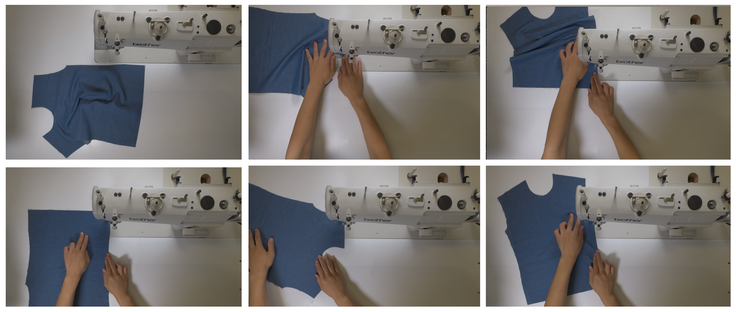
01 / TASK
Random-to-Target Flattening

RTFF asks a dual-arm robot to remove wrinkles and align a fabric panel to a user-specified flat pose at the same time.

Flattening changes pose. Pose correction can create new wrinkles.

- **R1** **Arbitrary target flat pose**
  Accept a user-specific target position and orientation, rather than a fixed pose.
- **R2** **Simultaneous flattening and pose alignment**
  Simultaneously flatten and align the fabric in a single unified framework.
- **R3** **Wrinkle-aware state estimation**
  Real-time access to local wrinkle states for flattening and the global fabric pose for alignment.

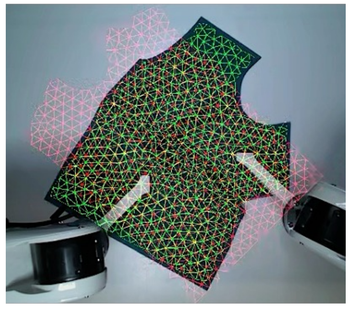
Efficient RTFF is important in fabric sewing, cutting, screen printing, and ironing, where flatness and precise alignment determine garment quality



02 / Representation
A shared geometric reference

Define both the current fabric state and the target state with the same *template mesh*, derived from the fabric's CAD model that deforms to match any observed state while preserving vertex identity.

Enable persistent vertex correspondence and direct per-vertex error .

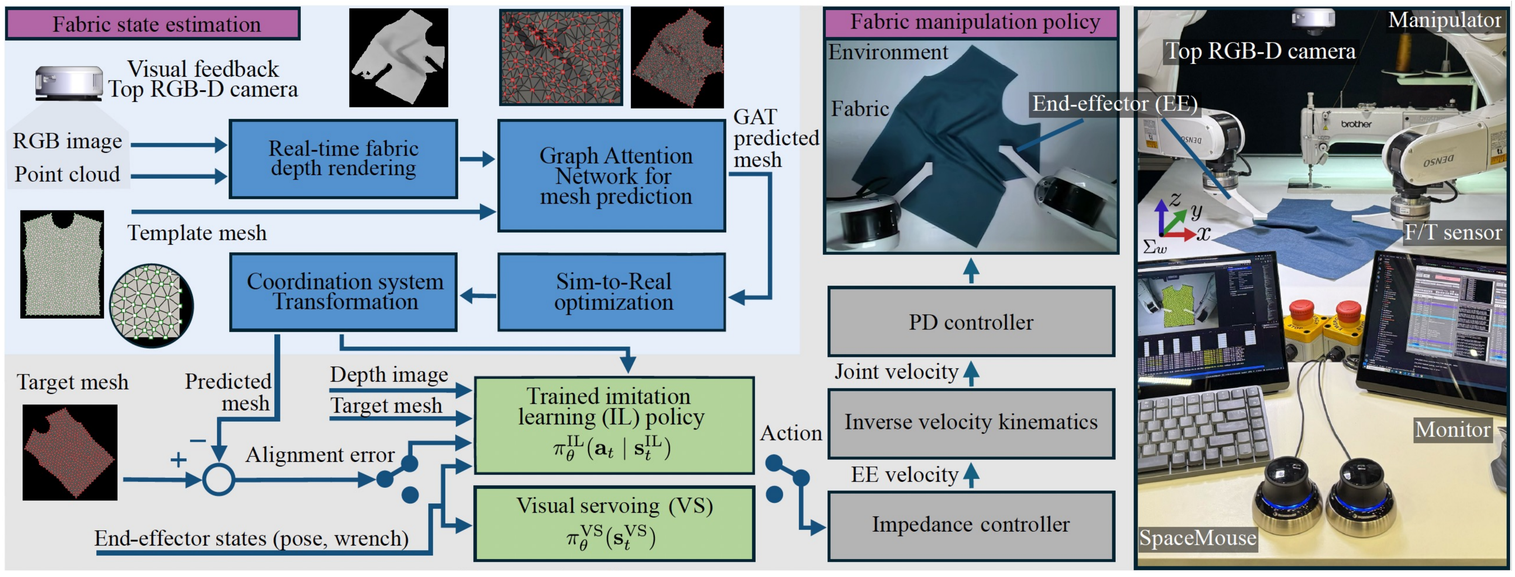


Let the canonical template mesh be $\boldsymbol{T} = (\boldsymbol{V}^{TPL}, \boldsymbol{E}^{TPL})$, a standardized flattened fabric reference derived from the fabric CAD model.
The current estimated mesh can then be represented as $\widehat{\boldsymbol{M}}_t = (\hat{\boldsymbol{V}}_t, \boldsymbol{E}^{TPL})$, and the target state is defined as $\boldsymbol{M}^* = (\boldsymbol{V}^*, \boldsymbol{E}^{TPL})$.

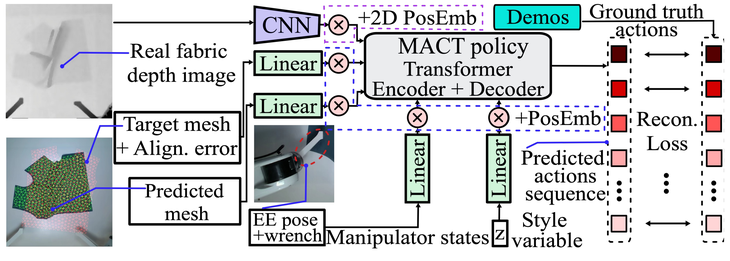
Using this representation we can define:
Per-vertex error: $\boldsymbol{e}_{i,t} = \hat{\boldsymbol{v}}_{i,t} - \boldsymbol{v}_i^* \in R^3$.
Mean alignment error: $\bar{\boldsymbol{e}}_t = \mathbf{1}/N \sum_i \boldsymbol{e}_{i,t}$.

03 / Methodology
The proposed RTFF policy framework

- **Template-mesh based fabric state estimation pipeline based on Graph Attention Network provides a common coordinate system.**
- **A Goal-conditioned Imitation learning policy, Mesh Action Chunking Transformer, achieves RTFF from a compact demonstration set.**
- **A hybrid IL-VS framework enable effective coarse-to-fine simultaneous fabric flattening and alignment with .**



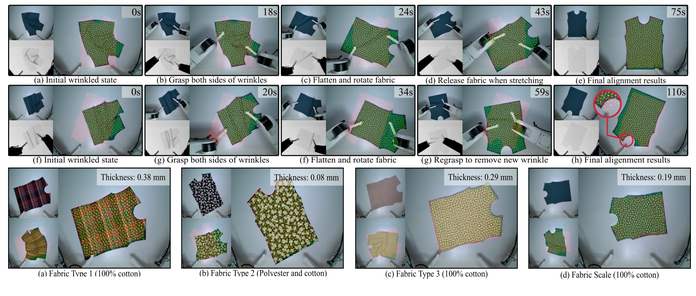
04 / Goal-conditioned Imitation Learning
Mesh Action Chunking Transformer



05 / Results
Evidence on real robot

- **The policy learn RTFF from 50 teleoperated episodes.**
- **Mesh state and alignment error are essential!**
- **Easy to generalize to fabric with different patterns and size.**

Method	Terminal alignment error (m)		IoU		Fail.
	Mean	Std.	Mean	Std.	
Ours	**0.011**	**0.002**	**0.926**	**0.018**	-
w/o VS (i)	0.028	0.010	0.871	0.05	-
w/o $\hat{M}_t$ and $\mathcal{E}_t$ (ii)	-	-	-	-	5/5
w D_t^{SEG}, w/o $\hat{M}_t$ and $\mathcal{E}_t$ (iii)	-	-	-	-	5/5
Type	0.0145	0.004	0.911	0.031	-
Scale	0.021	0.004	0.893	0.022	-



06 /
Key takeaway

- **K1** **Structured, unified representation is the foundation for robust manipulation.**
  A shared template mesh preserves correspondence between the current and target states, exposing both local wrinkles and global pose error. All in world coordinate system.
- **K2** **Explicit feedback makes imitation learning data-efficient.**
  Mesh-based state and alignment error provide goal-directed supervision, enabling strong policies from a compact set of demonstrations.
- **K3** **Learning–control handoff enables efficient and precise manipulation.**
  Learning transforms complex deformable states into a controllable low-dimensional regime, where visual servoing completes the final alignment.